

Transformation

In the context of image processing, "**transformations**" refer to operations that modify the spatial, color, or other properties of an image. These operations can be applied to the image to enhance its appearance, correct its orientation, change its perspective, or create various artistic effects. Transformations can be broadly categorized into two types: color transformations and spatial transformations.

Color Transformations:

Color transformations involve modifying the color properties of an image, such as its brightness, contrast, saturation, or hue. Common color transformations include:

- **Inverting:** Reversing the color values of the image, creating a “negative” of the original image. This can be used to create interesting visual effects or emphasize specific features of an image.
- **Adjusting brightness:** Changing the overall intensity of the image’s colors. This can be used to correct underexposed or overexposed images or create specific moods.
- **Adjusting contrast:** Modifying the difference between the darkest and lightest colors in the image. This can be used to enhance the details or create specific effects.
- **Adjusting saturation:** Changing the intensity of the colors in the image. This can be used to make the colors more vibrant or muted, or to create monochromatic or grayscale images.
- **Adjusting hue:** Shifting the colors in the image along the color spectrum. This can be used to change the overall color scheme of the image or create specific effects.

Spatial Transformations:

Spatial transformations involve changing the geometric properties of an image, such as its position, orientation, or size. Common spatial transformations include:

- **Flipping:** Mirroring the image either horizontally or vertically. This is useful for creating reflections or correcting image orientation.
- **Rotating:** Changing the orientation of the image by a specified angle. This can be used to correct image orientation or create dynamic compositions.
- **Scaling:** Changing the size of the image, either by enlarging or reducing it. This can be used to fit the image into a specific size or resolution.
- **Cropping:** Removing parts of the image outside a specified region. This can be used to focus on a specific area of the image or remove unwanted portions.
- **Warping:** Changing the perspective or shape of the image by applying a geometric transformation. This can be used to correct perspective distortion or create artistic effects.

These transformations can be applied individually or in combination to create a wide array of unique and visually engaging images. By using various image processing libraries like PIL, you can easily apply these transformations to your images and create custom effects tailored to your needs. We have more or less covered the color transformations in the previous lesson we will be focused on spacial transformation in this one.

Warping

True or False: Warping is a spatial transformation that can be used to correct perspective distortion or create artistic effects.

- ☒ True
- ☐ False

Check It!

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